

Vocabulary Work Bank KS1

<u>Year 1</u>		
<u>Computing systems and networks - Technology around us</u>	<u>Creating media - Digital painting</u>	<u>Creating media - Digital writing</u>
technology, computer, mouse, trackpad, keyboard, screen, double-click, typing.	paint program, tool, paintbrush, erase, fill, undo, shape tools, line tool, fill tool, undo tool, colour, brush style, brush size, pictures, painting, computers	word processor, keyboard, keys, letters, type, numbers, space, backspace, text cursor, capital letters, toolbar, bold, italic, underline, mouse, select, font, undo, redo, format, compare, typing, writing.
<u>Data and information – Grouping</u>	<u>Programming A - Moving a robot</u>	<u>Programming B – Programming animations</u>
object, label, group, search, image, property, colour, size, shape, value, data set, more, less, most, fewest, least, the same	Bee-Bot, forwards, backwards, turn, clear, go, commands, instructions, directions, left, right, route, plan, algorithm, program.	ScratchJr, command, sprite, compare, programming, area, block, joining, start, run, program, background, delete, reset, algorithm, predict, effect, change, value, instructions, design.

<u>Year 2</u>		
<u>Computing systems and networks - Information technology around us</u>	<u>Creating media - Digital music</u>	<u>Creating media - Digital photography</u>
<u>Information technology (IT), computer, barcode, scanner/scan</u>	<u>music, quiet, loud, feelings, emotions, pattern, rhythm, pulse, pitch, tempo, rhythm, notes, create, emotion, beat, instrument, open, edit.</u>	<u>device, camera, photograph, capture, image, digital, landscape, portrait, framing, subject, compose, light sources, flash, focus, background, editing, filter, format, framing, lighting,</u>
<u>Data and information – Pictograms</u>	<u>Programming A - Robot algorithms</u>	<u>Programming B - Programming quizzes</u>
<u>more than, less than, most, least, common, popular, organise, data, object, tally chart, votes, total, pictogram, enter, data, compare, objects, count, explain, attribute, group, same, different, conclusion, block diagram, sharing</u>	<u>instruction, sequence, clear, unambiguous, algorithm, program, order, prediction, artwork, design, route, mat, debugging, decomposition</u>	<u>sequence, command, program, run, start, outcome, predict, blocks, design, actions, sprite, project, modify, change, algorithm, build, match, compare, debug, features, evaluate, decomposition, code.</u>

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Article information:

Hybrid Optimization Algorithm in the Photometric Inversion of Periods for Asteroids -
IOPscience

<https://iopscience.iop.org/article/10.1088/1674-4527/acb9da/meta>

Article summary:

1. This article presents a hybrid optimization algorithm that combines the genetic algorithm with the Levenberg-Marquardt algorithm for the photometric inversion of rotational periods of asteroids.
2. The proposed hybrid algorithm is based on a Cellinoid shape model and significantly improves the efficiency of the inversion process.

3. Numerical experiments on synthetic lightcurves and sparse observations of real asteroids confirm that the proposed method performs well in improving computational efficiency.

Article rating:

Appears strongly imbalanced: The article is written in a biased or one-sided way, and the information it provides is not trustworthy enough to be considered a reliable source. You should consult other sources to find reliable information on the presented issues.

Article analysis:

The article titled "Hybrid Optimization Algorithm in the Photometric Inversion of Periods for Asteroids" discusses a new approach to improving the efficiency of inverting rotational periods of asteroids using a hybrid optimization algorithm. While the topic is interesting and relevant to the study of asteroids, there are several aspects of the article that need critical analysis.

Firstly, the article lacks a clear introduction or background information on why studying rotational periods of asteroids is important and how it contributes to our understanding of the solar system's evolution mechanism. This omission makes it difficult for readers who are not familiar with the subject to fully grasp the significance of the research.

Additionally, there is no mention of any potential biases or limitations in the methodology used. The article claims that the proposed hybrid algorithm improves computational efficiency, but there is no discussion on potential drawbacks or limitations of this approach. It would have been beneficial to include a comparison with other existing methods and their limitations to provide a more comprehensive analysis.

Furthermore, while the article mentions that multiple numerical experiments were performed on lightcurves and sparse observations of real asteroids, there is no detailed explanation or analysis of these experiments. Without this information, it is challenging to evaluate the reliability and validity of the results obtained using the proposed hybrid algorithm.

The article also lacks discussion on potential risks or uncertainties associated with using this hybrid optimization algorithm. It would have been valuable to address any potential sources of error or uncertainty in the inversion process and how they were mitigated or accounted for in this study.